Paper II – Urology (MCh Urology, Summer 2015)

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II-1. Describe medical management of BPH and role of various surgical modalities in its management.

Type: LAQ (25 marks)

Introduction and pathophysiology

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent) and a **dynamic** component (α1A-mediated smooth-muscle tone). The bladder responds with detrusor overactivity or later underactivity.

Evaluation

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention.
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

Medical management

Option	Key facts
Watchful waiting	Mild LUTS; lifestyle advice
α1-blockers (tamsulosin 0.4 mg, alfuzosin 10 mg, silodosin 8 mg)	First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery. Silodosin causes most anejaculation
5-ARIs (5α-reductase inhibitors) (finasteride 5 mg, dutasteride 0.5 mg)	For glands >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (maximal at 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down 51%
Combination	For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events
Antimuscarinic or β3-agonist (mirabegron)	Add for persisting storage symptoms; trials included only men with low PVR
Tadalafil 5 mg daily	LUTS with or without ED (erectile dysfunction); not with nitrates

Option	Key facts
Desmopressin	Nocturnal polyuria (night urine >33% of 24-h output); check sodium, especially ≥65 years

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.
- **AUR**: catheterise, then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines; α-blocker withdrawable after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy.

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection), monopolar or bipolar	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe when anticoagulation cannot stop; no histology
HoLEP/ThuLEP (laser enucleation)	Any size	Lowest retreatment, least bleeding
Simple prostatectomy (open or robotic)	>80 mL	Complete removal; diverticula or stones
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment
iTind (temporary implant)	25–75 mL, no median lobe	Removed after a few days; modest efficacy
PAE (prostate artery embolisation)	Larger glands (AUA 2023: may be offered)	Local anaesthesia; less effective than TURP; radiation, non-target embolisation

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).

- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI), then yearly.

Diagram to draw: Size-based surgical algorithm (≤30, 30–80, >80 mL).

> **Key points to remember:**
>
> - MTOPS: combination cut progression 66%; only 5-ARI-containing arms cut AUR and surgery.
> - AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
> - TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

II-2. Discuss metabolic evaluation in urinary calculus disease.

Type: LAQ (25 marks)

Introduction

- A chronic metabolic disease: up to 50% recur within 5–10 years. The ROKS (Recurrence of Kidney Stone) nomogram estimates individual risk.

Physicochemical basis

- Stones form from **supersaturated** urine: nucleation, growth, aggregation and retention (often on **Randall plaque**).
- In the **metastable zone** (between solubility and formation products) there is no spontaneous nucleation, but crystals grow; above it nucleation is spontaneous.
- **Inhibitors** (citrate, magnesium, Tamm-Horsfall protein) raise the supersaturation needed to crystallise.

Who needs evaluation

- **Screening evaluation** for every first-time former (AUA (American Urological Association), EAU (European Association of Urology)); **24-h urine** for high-risk and recurrent formers: children, family history, solitary kidney, uric acid, cystine or infection stones, GI (gastrointestinal) disease or bypass, hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, metabolic syndrome.

Screening evaluation

- **History**: diet, bowel disease, drugs (calcium, vitamins C and D, acetazolamide, topiramate, triamterene, indinavir).
- **NCCT (non-contrast computed tomography)**: burden; uric acid usually <500 HU (Hounsfield units).
- **Serum**: creatinine, electrolytes, calcium, uric acid, bicarbonate; PTH (parathyroid hormone) if calcium is high-normal.
- **Urine**: pH, crystals (hexagonal = cystine), culture; cyanide–nitroprusside screen for cystine, confirmed by quantitative 24-h cystine.
- **Stone analysis** (infrared, X-ray diffraction).

24-h urine

- One or two collections on usual diet, ≥4 weeks after the episode, stone-free and infection-free; adequacy by creatinine (men ~20–25, women ~15–20 mg/kg/day).

Parameter	Desirable per day	Abnormality and causes
Volume	**>2.5 L**	Low volume: commonest
Calcium	<250 mg women, <300 mg men, or <4 mg/kg	**Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism)
Oxalate	**<40 mg**	Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1))
Citrate	**>320 mg**	Distal RTA, diarrhoea, hypokalaemia, high animal protein
Uric acid	<600 mg (labs: <750 women, <800 men)	Hyperuricosuria promotes calcium oxalate stones

Parameter	Desirable per day	Abnormality and causes
pH	5.8–6.2	<5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection
Sodium	<100 mmol	Raises calcium excretion
Cystine	Negative screen	Cystinuria

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Distal RTA**: pH >6, low citrate, calcium phosphate stones. Incomplete forms: oral ammonium chloride load (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria**: autosomal recessive transport defect (cystine, ornithine, lysine, arginine); solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** hypomagnesiuria; RSS (relative supersaturation) guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT (adenine phosphoribosyltransferase) deficiency).

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Stone episode → screening evaluation
  Low risk → general advice
  High risk → 24-h urine → targeted therapy → repeat within 6 months → yearly
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Treatment linked to abnormality

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

Abnormality	Treatment
Hypercalciuria	**Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate
Primary hyperparathyroidism	**Parathyroidectomy**
Hypocitraturia, distal RTA	**Potassium citrate** 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; magnesium
Primary hyperoxaluria type 1	Pyridoxine, **lumasiran** (RNAi (ribonucleic acid interference)); liver ± kidney transplant
Hyperuricosuria	**Allopurinol** 300 mg/day
Uric acid stones	**Potassium citrate** to pH 6.0–6.5
Cystinuria	Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin**
Struvite	Complete clearance; AHA (acetohydroxamic acid) if surgery impossible

- **Borghi 1996**: urine >2 L cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

Special groups and follow-up

- **Children**: all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy**: defer 24-h urine (physiological hypercalciuria).
- Follow-up as in the flowchart, with blood tests for drug effects (potassium on thiazide or citrate); re-analyse new stones. Adherence is poor; citrus juices are only adjuncts.

Diagram to draw: Supersaturation zones; evaluation algorithm.

Key points to remember:

- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: NH4Cl fails to lower pH <5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-h urine within 6 months.

II-3a. Management of priapism

Type: SAQ (10 marks)

Definition and classification

- Full or partial erection **≥4 hours** beyond, or unrelated to, sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America)).
- **Ischaemic** (95%; veno-occlusive, rigid, painful, **emergency**); **non-ischaemic** (arterial fistula after trauma, partly rigid, painless); **stuttering** (recurrent ischaemic episodes, typically SCD (sickle cell disease)).
- **Causes**: ICI (intracavernosal injection), SCD (29–42% of males), α-blockers, trazodone, cocaine, leukaemia, metastases.

Diagnosis

- **Corporal blood gas**:

	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	**<30**	**>60**	**<7.25**
Mixed venous (flaccid)	40	50	7.35
Non-ischaemic (arterial)	>90	<40	7.40

- Colour Doppler if equivocal (absent flow or fistula); blood count, haemoglobin electrophoresis.

Ischaemic priapism

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Penile block → aspiration ± irrigation
→ Phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h
→ Distal shunt ± tunnelling → proximal shunt
→ >36 h or refractory → immediate prosthesis
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- Aspiration alone succeeds in under a quarter. Monitor blood pressure and pulse.

Shunt	Method
Winter (distal)	Biopsy needle through glans
Ebbehoj / T-shunt (distal)	Scalpel through glans; T-shunt turns blade 90°
Al-Ghorab (distal, open)	Excise tunica tips via dorsal glans incision
Snake tunnelling	Dilator passed proximally through distal shunt
Quackels (proximal)	Cavernosum to spongiosum at the bulb
Grayhack (proximal)	Cavernosum to saphenous vein
Penoscrotal decompression	Proximal corporotomy, clot evacuation

- **Duration predicts ED (erectile dysfunction)**: 0% ≤12 h, 22% at 12–24 h, 56% at 24–36 h, 100% >36 h.
- **Early prosthesis** gives less shortening than delayed insertion.
- Oral agents and hydration are adjuncts only; in SCD, treat intracavernosally with concurrent haematology care.

Non-ischaemic priapism

- Observe (about 60% resolve); reassess at 4 weeks. Then **selective embolisation**; repeat it, rather than surgery, for recurrence.

Stuttering priapism

- Prophylaxis: daily PDE5i (phosphodiesterase-5 inhibitor, which resets PDE5 regulation), pseudoephedrine, self-injected phenylephrine; hormonal agents (GnRH (gonadotropin-releasing hormone) agonists, antiandrogens) at the cost of libido.

Diagram to draw: Distal and proximal shunt sites.

> **Key points to remember:**
>
> - Ischaemic gas: pO2 <30, pCO2 >60, pH <7.25.
> - Distal shunt ± tunnelling before proximal; prosthesis if >36 h.
> - High-flow: observe (~60% resolve), then embolise.

II-3b. Renal tubular acidosis

Type: SAQ (10 marks)

Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular H+ secretion or HCO3 reabsorption.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	H+ secretion (α-intercalated cell)	Proximal HCO3 reabsorption	Aldosterone deficiency or resistance
Urine pH	**Always >6**	<5.5 at steady state	Low
Potassium	Low	Low	**High**
Serum HCO3	Low, may be very low	15–18 mEq/L	Mildly low
Urine calcium	High	Normal or mildly high	Low
Urine citrate	**Profoundly low**	Near normal	–
Stones	**Up to 70%; calcium phosphate, nephrocalcinosis**	Uncommon	Uncommon
Causes	Sjögren, lupus, inherited	Fanconi	Diabetes, obstruction
Treatment	**Potassium citrate** 1–2 mEq/kg/day	Alkali 10–15 mEq/kg/day plus potassium	Fludrocortisone, low-potassium diet, stop offending drugs

- **Why distal RTA forms stones**: fixed alkaline urine, profound hypocitraturia (acidosis increases citrate reabsorption) and hypercalciuria from bone buffering.
- **Incomplete distal RTA**: normal bicarbonate, but urine cannot be acidified below 5.5.

Diagnosis

- **UAG (urine anion gap) = (Na + K) – Cl** reflects urinary ammonium: negative in GI (gastrointestinal) loss, positive in renal causes.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): pH fails to fall below 5.5 in distal RTA. **Furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Fractional bicarbonate excretion** >15% during bicarbonate loading indicates proximal RTA.

Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim cause hyperkalaemic distal RTA; mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

Diagram to draw: Nephron with defect sites.

Key points to remember:

- Type 1: urine pH >6, hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause; NH4Cl load fails to lower pH <5.5 in distal RTA.

II-3c. Millin's prostatectomy

Type: SAQ (10 marks)

Definition

- **Retropubic extravesical simple prostatectomy** (Millin, 1945): enucleation through a transverse anterior capsulotomy without opening the bladder. The plane is the surgical capsule (compressed peripheral zone).

Indications and work-up

- LUTS (lower urinary tract symptoms) or obstructive complications with gland **>80–100 g**, or when lithotomy is impossible. Prefer suprapubic (Freyer) for a large median lobe, diverticulum or stones.
- Work-up: exclude cancer (PSA (prostate-specific antigen), MRI (magnetic resonance imaging)), TRUS (transrectal ultrasound) volume, cystoscopy, ± urodynamics.

Steps

1. Extraperitoneal approach to the **space of Retzius**; control the dorsal venous complex.
2. **Transverse capsulotomy ~2 cm below the bladder neck** (a longitudinal cut may extend into the sphincter).
3. **Finger enucleation** in the adenoma–capsule plane.
4. **Divide the apical urethra under vision**, sparing the sphincter.
5. Trim the bladder neck lip; advance mucosa into the fossa.
6. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
7. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Fossa haemostasis	Directly visible	Less visible
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	Dorsal venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%), urine leak, infection, ureteric injury, bladder neck contracture (2–6%), stricture, thromboembolism.
- Retrograde ejaculation 80–90%; ED (erectile dysfunction) 3–5%; incontinence rare.
- **Also know:** bleeding control by catheter traction, Malament purse-string or O'Conor capsular plication; vesicocapsular variant; versus TURP (transurethral resection of prostate): lower retreatment, no TUR syndrome, longer stay.

Modern alternatives

- HoLEP (holmium laser enucleation), robotic (multiport, single-port) prostatectomy: less bleeding.

Diagram to draw: Sagittal capsulotomy view.

Key points to remember:

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

II-3d. Energy sources in laparoscopic urology

Type: SAQ (10 marks)

Principles

- Devices seal by denaturing protein. Electrosurgery uses ~500 kHz current, too fast to stimulate nerve or muscle.
- **Cut**: continuous low-voltage waveform boils cell water. **Coagulation**: interrupted high-voltage waveform desiccates, spreading deeper. Also blend and spray (fulguration).

Types

Device	Principle	Lateral spread	Notes
Monopolar	Tip → patient → return pad	**5–7 mm**	Versatile; stray current
Bipolar	Current only between jaws	1–6 mm	No pad or capacitive coupling
Vessel sealer (LigaSure)	Compression plus impedance-controlled energy	1–3 mm	Seals vessels **≤7 mm**; single application
Ultrasonic (Harmonic)	Blade vibrating at 25–55 kHz; no current	**≈1 mm**	Safe with pacemakers; blade >200 °C, cools slowly
Argon beam	Monopolar current through argon, non-contact	Superficial	Raw surfaces; **gas embolism**

- **Also know:** lasers (CO2, KTP (potassium titanyl phosphate), holmium), now rarely used; laparoscopic RFA (radiofrequency ablation, target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses.

Electrosurgical injury

- Unseen injuries present late as perforation.

1. **Insulation failure**.
2. **Direct coupling**: active electrode touches another metal instrument.
3. **Capacitive coupling**: current induced through intact insulation; avoid hybrid trocars (metal cannula, plastic anchor).
4. **Thermal spread** to ureter, nerves or bowel.
5. **Return pad burns**; 6. **surgical fire**.

Safety

- **AEM (active electrode monitoring)** detects insulation failure and capacitive coupling; **REM (return electrode monitoring)** detects poor pad contact.
- Lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view, inspect insulation; bipolar near vital structures and in pacemaker patients.

Diagram to draw: Monopolar vs bipolar circuit.

Key points to remember:

- Spread: monopolar 5–7 mm > bipolar 1–6 > LigaSure 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz.
- Stray current: insulation failure, direct and capacitive coupling; use AEM and REM.

II-3e. Role of vascular grafts in urology

Type: SAQ (10 marks)

Introduction

- Grafts and patches replace, bypass, extend or widen vessels when primary repair fails or narrows them.

Types

Type	Examples	Pros	Cons
Autologous vein	Saphenous, gonadal	Resists infection; small vessels	Limited length
Autologous artery	Internal iliac, Carrel aortic patch	Renal arteries	Limited supply
Biological	Bovine pericardium	Pliable patch	Cost
Synthetic	**PTFE (polytetrafluoroethylene)**, Dacron	Off-the-shelf, large calibre	Infection; venous grafts thrombose

Applications

1. **Renal transplantation**: multiple arteries (18–43%) joined by Carrel patch or bench reconstruction, preserving the lower pole artery (supplies the ureter). Short right renal vein extended with donor IVC (inferior vena cava), external iliac or spiral gonadal vein. Transplant artery stenosis: angioplasty first, graft if it fails.
2. **RCC (renal cell carcinoma) with IVC thrombus**: patch cavoplasty if residual lumen would be <50%; **PTFE replacement (16–20 mm)** if the wall is invaded circumferentially.
3. **Post-chemotherapy RPLND (retroperitoneal lymph node dissection)**: IVC: repair if narrowing <25%, patch or graft if more; ligation tolerated with collaterals. Aorta: about 1% need PTFE or Dacron replacement; never ligated.
4. **Renovascular surgery and autotransplantation**: aortorenal bypass (saphenous vein, internal iliac artery, PTFE), splenorenal or hepatorenal bypass, ex vivo bench repair of branch disease.
5. **Dialysis access**: PTFE AV (arteriovenous) graft.

- **Also know:** iatrogenic or traumatic vascular injury (interposition graft); retroperitoneal sarcoma caval resection; superior mesenteric vein injury (vein graft); bland infrarenal thrombus (filter or caval interruption).

Principles

- Autologous graft in contaminated fields and small vessels; synthetic for large high-flow vessels.
- Tension-free, spatulated anastomosis with fine monofilament (polypropylene); heparin before clamping; antiplatelet or anticoagulant afterwards as indicated; omental wrap separates synthetic grafts from bowel and urine.

Complications

- Thrombosis (caval grafts 10–40%), stenosis, infection (often needing removal), enteric or urinary fistula.

Diagram to draw: Bench arterial reconstruction.

Key points to remember:

- Transplant: Carrel patch, bench reconstruction; preserve the lower pole artery.
- IVC: patch if lumen <50%; PTFE replacement if wall invaded.
- Post-chemotherapy RPLND: IVC can be ligated, aorta must be grafted.

II-3f. Transrectal ultrasonography

Type: SAQ (10 marks)

Physics and technique

- Piezoelectric crystals; higher frequency: better resolution, less penetration.
- **6–10 MHz** end-fire, side-fire or biplane probes; left lateral position after DRE (digital rectal examination).

Sonographic anatomy

Zone (McNeal)	Glandular volume	TRUS appearance	Cancers
Peripheral	~70%	Homogeneous, echogenic	~70%
Transition	5–10% (BPH (benign prostatic hyperplasia))	Heterogeneous	~20–25%
Central	~25%	Homogeneous, echogenic	~5–10%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Corpora amylacea mark the surgical capsule; seminal vesicles about 4.5–5.5 × 2 cm.

Measurements

- **Volume (ellipsoid) = 0.52 × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. Planimetry for brachytherapy.
- **PSAD (PSA (prostate-specific antigen) density)** = PSA/volume; >0.15 is suspicious.

Findings

- **Cancer**: classically hypoechoic peripheral-zone lesion, but 39% are isoechoic; false positives are BPH nodules and prostatitis.
- **Cysts**: midline Müllerian; off-midline ejaculatory duct cyst.

Indications

- Biopsy; volume before BPH surgery or brachytherapy; ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst); abscess drainage.

TRUS (transrectal ultrasound)-guided biopsy

- **12-core** systematic biopsy: medial and lateral cores at base, mid-gland and apex (sextant alone is inadequate). Periprostatic lidocaine block at the junction of prostate base and seminal vesicle.
- Contraindications: coagulopathy, acute prostatitis, severe immunosuppression. Continue low-dose aspirin; stop warfarin or clopidogrel 5–7 days.
- Complications: haematuria, haematospermia, rectal bleeding, **sepsis** (0.3–3%) from FQ (fluoroquinolone)-resistant E. coli; povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.

Current shift

- **Transperineal biopsy** preferred by EAU (European Association of Urology) for lower infection risk.

- **MRI (magnetic resonance imaging)-first pathway**: PROMIS (2017): mpMRI (multiparametric MRI) sensitivity 93% versus 48% for TRUS biopsy. PRECISION (2018): MRI-targeted biopsy found more significant cancer (38% vs 26%). Targeting: cognitive, fusion or in-bore.
- **Also know:** colour Doppler, contrast-enhanced TRUS, elastography; micro-ultrasound (29 MHz) comparable to mpMRI; saturation biopsy after negative biopsies; treated glands become diffusely hypoechoic.

Diagram to draw: 12-core biopsy template.

> **Key points to remember:**
> - Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
> - 39% of cancers isoechoic; 12-core biopsy; sepsis 0.3–3%.
> - PROMIS and PRECISION established the MRI-first pathway.

Rapid Revision Sheet – Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. Describe medical management of BPH and role of various surgical modalities in its management. (LAQ)

- MTOPS: combination cut progression 66%; only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

II-2. Discuss metabolic evaluation in urinary calculus disease. (LAQ)

- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: NH4Cl fails to lower pH <5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-h urine within 6 months.

II-3a. Management of priapism (SAQ)

- Ischaemic gas: pO2 <30, pCO2 >60, pH <7.25.
- Distal shunt ± tunnelling before proximal; prosthesis if >36 h.
- High-flow: observe (~60% resolve), then embolise.

II-3b. Renal tubular acidosis (SAQ)

- Type 1: urine pH >6, hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause; NH4Cl load fails to lower pH <5.5 in distal RTA.

II-3c. Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

II-3d. Energy sources in laparoscopic urology (SAQ)

- Spread: monopolar 5–7 mm > bipolar 1–6 > LigaSure 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz.
- Stray current: insulation failure, direct and capacitive coupling; use AEM and REM.

II-3e. Role of vascular grafts in urology (SAQ)

- Transplant: Carrel patch, bench reconstruction; preserve the lower pole artery.
- IVC: patch if lumen <50%; PTFE replacement if wall invaded.
- Post-chemotherapy RPLND: IVC can be ligated, aorta must be grafted.

II-3f. Transrectal ultrasonography (SAQ)

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; 12-core biopsy; sepsis 0.3–3%.
- PROMIS and PRECISION established the MRI-first pathway.